Assignment: Object Cloning in JavaScript

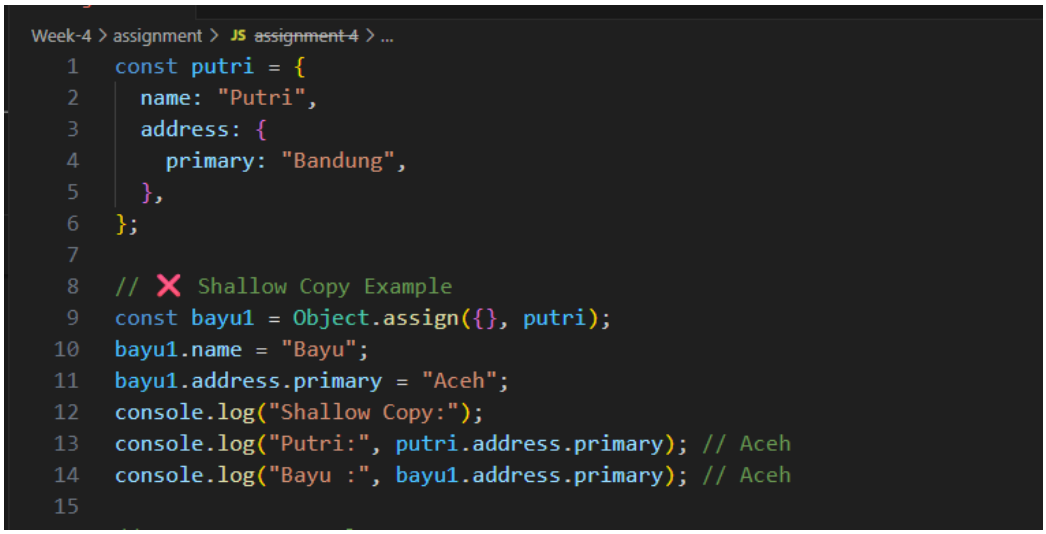
Question

Why is Putri's primary address showing as Aceh even after changing Bayu's address, and how can we fix it so that Bayu's address changes to Aceh while Putri's stays Bandung?

Answer

1. Explanation

When we assign one object to another (like cloning 'putri' into 'bayu'), JavaScript performs a shallow copy. A shallow copy means the nested objects (like 'address') are shared between both variables. So, when you modify bayu.address.city, it also changes putri.address.city because both point to the same memory location.



Output:



2. How to Fix It (Deep Copy)

To make each object completely independent, we must create a deep copy.

```
// Deep copy
// Deep copy section
const putriDeep = {
  name: "Putri",
  birthday: "2005-02-13",
  phone: "081234567890",
  address: {
    primary: "Bandung",
    secondary: "Sukabumi",
  },
};

const bayuDeep = structuredClone(putriDeep);

bayuDeep.name = "Bayu";
bayuDeep.address.primary = "Aceh";

console.log("\nDeep Copy:");
console.log("Putri:", putriDeep.address.primary);
console.log("Bayu :", bayuDeep.address.primary);
```

Output:

```
>> node assignment4.js
>>
Shallow Copy:
Putri: Aceh
Bayu : Aceh

Deep Copy:
Putri: Bandung
Bayu : Aceh
```